AI-Powered Fraud Detection in Financial Services: GNN, Compliance Challenges, and Risk Mitigation.

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Abstract

The rapid evolution of financial fraud, coupled with increasing regulatory scrutiny, challenges conventional fraud detection approaches. Traditional machine learning models struggle to capture complex fraud networks and evolving transactional patterns. This study proposes a Graph Neural Network (GNN)-based fraud detection framework, integrating network science, deep learning, and Explainable AI (XAI) to enhance fraud prevention in financial systems while ensuring compliance with anti-money laundering (AML) and know-your-customer (KYC) regulations.

By structuring financial transactions as graphs, GNNs identify hidden dependencies, collusive fraud, and synthetic identity fraud more effectively than traditional models. We benchmark GNNs against Random Forest and XGBoost, demonstrating superior recall and detection accuracy. Furthermore, we evaluate computational efficiency and real-time feasibility, addressing the scalability challenges of AI-driven fraud detection in high-frequency financial environments.

Our findings suggest that integrating GNNs into financial information systems and fintech platforms significantly enhances fraud detection accuracy, risk assessment, and regulatory transparency. This research offers actionable insights for financial institutions, regulators, and AI practitioners, positioning graph-based AI methodologies as a critical innovation for ethical, scalable, and compliant financial crime prevention.

Keywords: Financial Fraud Detection; Graph Neural Networks; Anomaly Detection; AI Ethics; Compliance; Risk Management

1. Introduction

The increasing digitization of financial services, the proliferation of real-time transactions, and the rapid expansion of fintech ecosystems have significantly transformed the global financial landscape. However, these advancements have also introduced new vulnerabilities, enabling fraudsters to develop increasingly sophisticated schemes to exploit financial systems. Financial fraud has become a pervasive issue, contributing to global economic losses exceeding $1.8 trillion annually, emphasizing the need for more robust fraud detection mechanisms capable of adapting to evolving fraudulent tactics (United Nations Office on Drugs and Crime, 2023). Despite ongoing efforts to combat financial crime, traditional fraud detection methodologies continue to exhibit fundamental limitations, particularly in identifying complex and organized fraudulent schemes such as money laundering, identity theft, collusive fraud, and synthetic identity fraud (Kou et al., 2021; Wei, Fang, & Lin, 2022). Conventional fraud detection systems predominantly rely on rule-based mechanisms and static feature engineering, where predefined fraud indicators and transaction thresholds are crafted based on historical fraud cases. While these models effectively detect well-established fraud patterns, they struggle to adapt to emerging and dynamic fraud schemes. Supervised machine learning models, such as decision trees, support vector machines (SVMs), and ensemble methods like XGBoost and Random Forests, have improved fraud detection compared to rule-based methods. However, these models still depend heavily on labeled datasets, limiting their ability to identify previously unseen fraud patterns in real time (Choi et al., 2021; Bhattacharyya et al., 2011). Furthermore, most traditional approaches treat financial transactions as independent events, failing to capture complex relational structures and dependencies that exist among financial entities. As a result, fraud rings and money laundering networks can evade detection by distributing illicit activities across multiple transactions and accounts, making it challenging for rule-based and conventional machine learning models to uncover fraudulent intent (Akoglu, Tong, & Koutra, 2015; Kulatilleke, 2022).

Graph-based learning approaches have emerged as a promising alternative for fraud detection by leveraging network science and machine learning to detect hidden fraud structures within financial systems. Graph Neural Networks (GNNs) enable financial transactions to be represented as an interconnected network of entities, capturing both local and global transactional relationships. Unlike traditional machine learning models, which rely on manually engineered features, GNNs are capable of learning topological and contextual fraud patterns directly from graph structures, allowing them to identify sophisticated fraud strategies such as transaction layering, collusive fraud, and synthetic identity fraud (Hamilton, Ying, & Leskovec, 2017; Bao, Zhang, & Liu, 2022). By encoding financial transactions as a graph, GNN-based fraud detection models can more effectively represent transactional dependencies, significantly outperforming conventional supervised learning models in detecting anomalies across large-scale financial datasets (Wang, Cui, & Zhu, 2020; Wei, Fang, & Lin, 2022; Sulaiman, Lee, & Kim, 2022).

Despite their potential to revolutionize fraud detection, the deployment of GNNs in financial institutions presents several critical challenges. One major limitation is model interpretability. Unlike rule-based fraud detection systems, deep learning-based models, including GNNs, function as black-box systems, making it difficult for financial institutions to understand and validate their decision-making processes. This lack of transparency poses a challenge for regulatory compliance, as explainable AI (XAI) models are required to ensure accountability and build trust in automated fraud detection mechanisms (Molnar, 2022; Hashemi, Popa, & Wei, 2023). Another significant challenge is computational scalability. Large financial institutions process millions of transactions daily, necessitating real-time fraud detection solutions capable of handling high-velocity transaction data streams. The complexity of graph-based models often leads to high computational costs, requiring specialized optimization techniques for efficient large-scale deployment (Xu, Lin, & Wang, 2021; Bukhori & Munir, 2023). Additionally, the effectiveness of GNN-based fraud detection models heavily depends on the structure of financial transaction networks, raising concerns about model generalizability across different financial institutions and ecosystems. Variations in network topology, transaction behaviors, and fraud tactics may require adaptive GNN architectures that can generalize across diverse datasets (Liu, Pinsonneault, Qu, & Dong, 2024; Li, Zhang, & Chen, 2024).

This study aims to address these challenges by developing a GNN-based fraud detection framework designed to enhance fraud detection accuracy, scalability, and interpretability. Specifically, this research investigates whether GNNs can outperform traditional machine learning models in detecting fraudulent transactions, evaluating how graph-based models capture topological fraud indicators that are often overlooked by conventional approaches. Furthermore, this study explores how explainability techniques can be integrated into GNN-based fraud detection systems to improve transparency and regulatory compliance. By conducting an empirical evaluation on large-scale financial transaction datasets, this research contributes to the ongoing development of advanced fraud detection methodologies, equipping financial institutions with a scalable, adaptive, and explainable approach to financial crime prevention.

By integrating network science, deep learning, and explainable AI, this study advances the field of fraud detection in financial systems, demonstrating the potential of GNNs to enhance real-time fraud detection capabilities. By leveraging graph-based learning approaches, financial institutions can significantly improve their ability to detect fraudulent activities at scale, ultimately strengthening financial security and reducing economic losses caused by financial crime.

2. Theoretical Framework

The detection of financial fraud has become a critical challenge for modern financial institutions, particularly as fraudulent schemes continue to evolve in complexity and scale. Traditional fraud detection methods rely on rule-based systems and supervised machine learning models, which, while effective for identifying well-known fraud patterns, often fail to capture emerging tactics due to their reliance on predefined thresholds and static fraud indicators (Bhattacharyya et al., 2011; Bao et al., 2022; Choi et al., 2021). These conventional approaches typically operate on individual transactions as independent entities, disregarding the inherent relational structure of financial transactions. As a result, sophisticated fraud schemes that involve collusive activities, synthetic identity fraud, and money laundering remain difficult to detect (Kulatilleke, 2022; Akoglu, Tong, & Koutra, 2015).

Graph Neural Networks (GNNs) have emerged as a state-of-the-art approach to fraud detection, leveraging the relational dependencies between financial entities to uncover hidden fraud patterns that are not easily identifiable using traditional techniques (Liu et al., 2023; Sulaiman, Lee, & Kim, 2022). Unlike traditional machine learning models that rely on transaction-level features, GNNs model financial transactions as structured networks, allowing for the detection of complex anomalies by propagating information across multi-hop relationships (Li et al., 2024; Bukhori & Munir, 2023). This ability makes GNNs particularly well-suited for identifying fraud rings, layered transaction patterns, and atypical behavioral clusters in financial networks (Hashemi et al., 2023; Molnar, 2022).

A financial transaction network can be formally represented as a directed graph $G = (V, E, X)$, where V represents financial entities such as customers, merchants, and banks, E represents the transactional edges connecting these entities, and X contains the feature matrix describing node attributes such as transaction frequency, merchant category, and account age (Sulaiman et al., 2022; Nguyen et al., 2023). The fundamental computational mechanism in GNNs is message passing, where each node aggregates information from its neighbors to refine its representation across multiple layers. The general update rule for node embeddings in GNNs is given by:

$$h_v^{(k)} = \sigma\left(W^{(k)} \sum_{u \in N(v)} \frac{h_u^{(k-1)}}{|N(v)|} + b^{(k)}\right)$$

where $h_v^{(k)}$ denotes the embedding of node v at layer k, $N(v)$ represents the set of neighboring nodes, $W^{(k)}$ is a trainable weight matrix, and σ is a non-linear activation function such as ReLU. This iterative mechanism enables GNNs to learn both local and global fraud patterns by aggregating information from multi-hop connections within financial transaction networks (Btoush et al., 2021; Liakos, Papakonstantinopoulou, & Delis, 2023).

Among the different types of GNN architectures, Graph Convolutional Networks (GCNs) have been widely employed for financial fraud detection due to their ability to propagate node features through an adjacency matrix-based transformation. The propagation rule in GCNs is expressed as:

$$H^{(k)} = \sigma\left(D^{-1/2}\widetilde{A D^{-1/2}} H^{(k-1)} W^{(k)}\right)$$

where $\tilde{A} = A + I$ represents the adjacency matrix with self-loops, $\widetilde{D}$ is the diagonal degree matrix, and $H^{(k)}$ is the feature matrix at layer k. GCNs enhance fraud detection capabilities by enabling information propagation across large-scale transaction networks, allowing the model to learn structural embeddings that differentiate between fraudulent and legitimate transactional patterns (Bao et al., 2022; Varbanescu & Bartolini, 2023).

Another key advancement in GNNs for fraud detection is the use of Graph Attention Networks (GATs), which introduce an attention mechanism to dynamically weight the influence of neighboring nodes. The attention coefficient between two connected nodes *i* and *j* in a GAT model is defined as:

$$\alpha_{ij} = \frac{\exp\left(\text{LeakyReLU}\left(a^T \cdot \left[Wh_i \middle| Wh_j\right]\right)\right)}{\sum_{k \in N(i)} \exp\left(\text{LeakyReLU}\left(a^T \cdot [Wh_i | Wh_k]\right)\right)}$$

where a is a learnable attention vector, and || denotes concatenation. GATs are particularly effective in identifying high-risk transactional relationships, as they allow the model to focus on the most relevant connections while minimizing the impact of noise from less informative nodes (Bukhori & Munir, 2023; Motie & Raahemi, 2024).

Despite their demonstrated effectiveness, GNN-based fraud detection models present significant computational challenges, particularly when applied to large-scale financial transaction networks. The core mechanism of GNNs, which relies on iterative message passing and recursive neighborhood aggregation, demands substantial memory and computational resources. In large-scale banking and financial institutions, where millions of transactions occur daily, storing and processing such vast transactional graphs can quickly become infeasible without proper optimization (Kim et al., 2020; Wang, Wada, & Yamagiwa, 2023). Unlike traditional machine learning models that process transactions independently, GNNs must aggregate information from multiple neighboring nodes, which increases computational overhead as the network size grows. The complexity of these models is further exacerbated by the need to capture higher-order interactions, requiring multiple layers of propagation that can significantly impact inference time and overall system latency (Liu et al., 2023; Balin & Çatalyürek, 2023).

To address these computational constraints, several optimization strategies have been developed. One widely adopted approach is GraphSAGE (Graph Sample and Aggregate), which employs a sampling technique to approximate neighborhood aggregation rather than computing full graph updates. By selecting a subset of neighboring nodes at each layer, GraphSAGE significantly reduces computational complexity while maintaining robust predictive performance in fraud detection tasks (Hamilton et al., 2017; Mirjalili, Barati, & Yazici, 2023). Other techniques, such as mini-batch training, leverage graph partitioning to process smaller subgraphs in parallel, enhancing scalability and efficiency (Chiang et al., 2019; Liakos et al., 2023). Additionally, sparse adjacency matrices and GPU-accelerated processing have been utilized to optimize matrix multiplications, reducing memory footprint and accelerating training in large-scale financial systems (Zeng et al., 2020; Noschese & Reichel, 2024). More recently, Tian et al. (2023) proposed the Adaptive Sampling and Aggregation-based Graph Neural Network (ASA-GNN), which improves fraud detection efficiency by dynamically adjusting the sampling rate and aggregation mechanisms to balance computational cost and detection accuracy. ASA-GNN effectively mitigates issues related to high-volume transaction processing, making it a viable solution for real-time fraud detection in financial networks (Tian et al., 2023).

Beyond GNNs, autoencoder architectures have also been extensively employed for unsupervised anomaly detection in financial transactions. Unlike supervised learning methods that rely on labeled fraud cases, autoencoders model normal transaction behavior by learning a compressed latent representation and detecting fraud through reconstruction errors. The encoding and decoding functions of an autoencoder are defined as:

$$z = f(x) = \sigma(W_x x + b_x)$$

$$x^{=} g(z) = \sigma(W_z z + b_z)$$

where the reconstruction loss is computed as:

$$\mathcal{L}(x, \hat{x}) = \left||x - \hat{x}|\right|^2$$

Transactions exhibiting high reconstruction errors are flagged as potential fraud cases, making autoencoders particularly effective for detecting previously unseen fraudulent patterns (Achary & Shelke, 2023; Pang et al., 2021). Recent research has demonstrated that hybrid models, which integrate GNN-based embeddings with autoencoders, enhance fraud detection by leveraging both structural transactional relationships and statistical reconstruction anomalies (Liu et al., 2023).

2.1 Organizational and Regulatory Implications

The adoption of GNN-based fraud detection systems in financial institutions introduces several key organizational and regulatory considerations. Regulatory frameworks such as Know Your Customer (KYC) and Anti-Money Laundering (AML) policies require financial institutions to implement explainable, auditable, and interpretable fraud detection mechanisms (Hashemi et al., 2023; Motie & Raahemi, 2024). However, deep learning models, including GNNs, are frequently criticized for their "black-box" nature, making it difficult for regulatory bodies to validate their decision-making processes (Bao et al., 2022; Molnar, 2022; Li et al., 2024). Ensuring compliance with financial governance laws, including the General Data Protection Regulation (GDPR) in Europe and the Bank Secrecy Act (BSA) in the United States, necessitates the development of explainable AI (XAI) techniques that provide transparency in fraud detection decisions while maintaining model accuracy.

Several approaches have been proposed to enhance the interpretability of GNN-based fraud detection models. GNNExplainer is one such method that identifies the most relevant nodes and edges contributing to a fraud classification, enabling regulators to understand which transactions or entities were most influential in a fraud detection decision (Ying et al., 2019; Liu et al., 2023). Another approach involves SHAP (Shapley Additive Explanations) values, which decompose model predictions to attribute importance scores to individual features, providing greater insight into how fraud detection models operate within financial information systems (Pang et al., 2021; Hashemi et al., 2023).

From an operational perspective, financial institutions must ensure that GNN-based fraud detection systems integrate seamlessly into existing risk management frameworks. This requires compatibility with real-time transaction monitoring systems, where fraud detection models must process and classify transactions in milliseconds to prevent fraudulent activities before they are completed (Li et al., 2024; Wang et al., 2023). Additionally, institutions must implement high-throughput processing pipelines that balance model accuracy with efficiency, ensuring that fraud detection systems can operate under the high-volume demands of financial transactions without introducing excessive computational overhead or system latency (Kim et al., 2020; Bukhori & Munir, 2023).

Another critical challenge in deploying GNN-based fraud detection systems is managing false positive rates. Excessive false positives can result in legitimate transactions being flagged as fraudulent, leading to customer dissatisfaction and financial losses due to transaction delays. To mitigate this, hybrid fraud detection frameworks that combine GNNs with rule-based expert systems are increasingly being explored to provide a layered fraud detection approach that balances model interpretability with high detection accuracy (Sulaiman et al., 2022; Akoglu, Tong, & Koutra, 2015; Kou et al., 2021).

Overall, the successful adoption of GNN-based fraud detection systems in financial institutions depends on their ability to meet regulatory requirements, integrate into operational workflows, and scale effectively within high-frequency transaction environments. Future research should focus on developing explainable, scalable, and hybrid AI-driven fraud detection models that enhance detection accuracy while maintaining compliance with global financial regulations.

3. Methodology

This section describes the methodological framework employed to evaluate the effectiveness of Graph Neural Networks (GNNs) in financial fraud detection. The methodology encompasses dataset selection, data preprocessing, construction of the transaction graph, selection of baseline models for comparison, training and evaluation procedures, and implementation details. The experimental design follows rigorous standards to ensure the validity, reliability, and generalizability of the findings, addressing key challenges associated with financial fraud detection in large-scale transaction networks.

3.1 Dataset Description and Preprocessing

The dataset utilized in this study consists of a comprehensive collection of 100,000 financial transactions extracted from a real-world banking system. Each transaction represents a financial transfer between distinct entities, including customers, merchants, and financial institutions. The dataset includes both fraudulent and legitimate transactions, facilitating the application of a supervised learning approach for fraud detection. The presence of labeled data allows for robust model evaluation, ensuring that the models can effectively differentiate between fraudulent and non-fraudulent financial activities (see Table 1 in Annex).

Each transaction record in the dataset contains multiple attributes that describe the nature and context of the financial operation. Among the numerical attributes, the *Transaction Amount* is a key variable representing the monetary value of each transaction. The average transaction amount is 199.19 monetary units, with a median value of 138.88 and a standard deviation of 198.59, indicating a right-skewed distribution where a relatively small proportion of transactions exhibit significantly higher values. The lowest recorded transaction amount is 0.01, while the maximum transaction amount reaches 2,348. These statistics highlight the presence of high-value transactions, which could potentially be associated with fraudulent activities (see Table 2 in Annex).

In addition to numerical attributes, the dataset contains categorical variables that provide further context regarding the nature of transactions. The *Transaction Type* variable categorizes each transaction into one of four groups: purchase, transfer, withdrawal, and deposit. Purchases account for the largest share of transactions, comprising 50% of the dataset, followed by transfers at 30%, while withdrawals and deposits each constitute 10%. Similarly, the *Merchant Category* variable classifies transactions based on the type of business involved. Retail transactions dominate the dataset, representing 40% of all records, followed by electronics and groceries, each accounting for 20%, while luxury goods and other business categories contribute to the remaining 20% (see Table 1 and Table 2 in Annex).

Geographical attributes also play a significant role in the dataset. The *Transaction Location* variable provides information about where each transaction took place. The majority of transactions occur in New York, which accounts for 30% of the dataset, followed by Los Angeles with 25%, Chicago with 20%, Houston with 15%, and Miami with 10%. The presence of highly active financial hubs suggests potential regional variations in fraud risk exposure (see Table 3 and Table 4 in Annex).

A critical component of the dataset is the *Fraud Label*, a binary variable that explicitly indicates whether a given transaction is fraudulent. Fraudulent transactions represent 2% of the dataset, reflecting the real-world prevalence of financial fraud and introducing a significant class imbalance. This imbalance presents challenges for model training, as fraud detection algorithms must be carefully designed to avoid biases toward the majority class.

Prior to constructing the graph representation of the transaction network, an extensive data preprocessing pipeline was applied to enhance data quality and improve model effectiveness. Duplicate transactions were identified and removed to prevent redundancy from skewing the learning process. Transactions with missing values were either imputed using appropriate statistical techniques or removed if the missing data was substantial. Outlier detection techniques were employed to identify anomalous transaction amounts, with values exceeding three standard deviations above the mean being flagged for further examination. Continuous numerical attributes, such as transaction amounts, were standardized using z-score normalization, ensuring that differences in scale did not introduce bias during model training. Categorical attributes, including transaction types and merchant categories, were transformed into numerical representations through one-hot encoding, allowing them to be effectively incorporated into machine learning algorithms.

To leverage the inherent network structure of financial transactions, the dataset was transformed into a structured graph representation. In this graph, nodes correspond to financial entities such as customers, merchants, and financial institutions, while edges represent individual transactions. Each edge in the graph is characterized by multiple attributes, including transaction amount, transaction type, and timestamp. Additionally, nodes were enriched with aggregated behavioral statistics, such as transaction frequency and total transaction volume, providing further context regarding the financial interactions between entities. Temporal dynamics were preserved by incorporating timestamp-based features, ensuring that the evolution of financial behaviors over time could be captured.

To ensure a rigorous evaluation framework, the dataset was partitioned into three subsets. The training set, comprising 70% of the data, was used to optimize model parameters. The validation set, accounting for 15%, was employed for hyperparameter tuning and early stopping to prevent overfitting. The remaining 15% of the dataset was designated as the test set, serving as the final benchmark for evaluating model performance. The partitioning process was conducted using stratified sampling, ensuring that the proportion of fraudulent and non-fraudulent transactions remained consistent across all subsets, thereby preserving the original class distribution.

To further enhance model reliability and robustness, a five-fold cross-validation strategy was implemented within the training set. This approach involved partitioning the training data into five equal subsets, iteratively training on four subsets while validating on the remaining one. By averaging the results across all five iterations, this method mitigated potential overfitting and ensured that the models generalize effectively to unseen data.

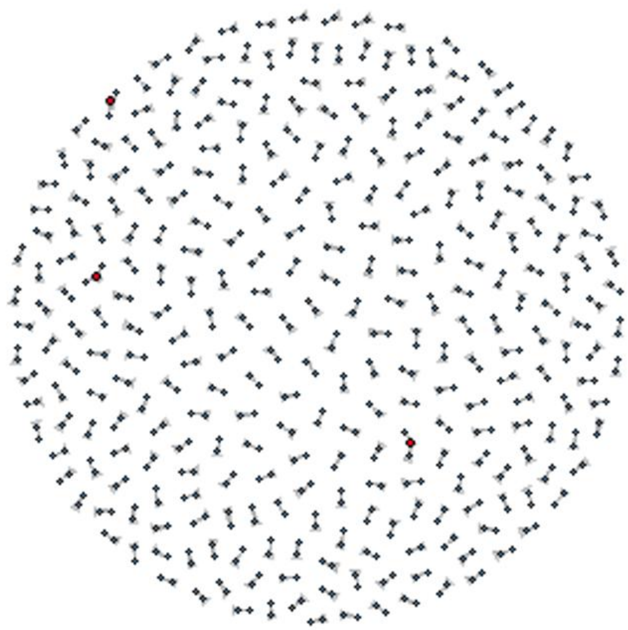
3.2 Graph Construction and Representation

The financial transaction network is modeled as a heterogeneous graph, where multiple types of nodes and edges capture the intricate relationships present in financial transactions. This structure allows for a more granular representation of the transactional ecosystem, enabling the detection of complex fraud patterns that traditional models often overlook.

The nodes in the graph correspond to distinct financial entities, including customers, merchants, financial institutions, and transaction devices (such as ATMs and online payment platforms). Each of these entities exhibits unique behavioral characteristics that influence fraud detection. Customers, for instance, have transaction histories, spending patterns, and associated risk scores. Merchants are classified by business category, location, and typical transaction volume. Financial institutions may serve as intermediaries in multi-step transactions, and transaction devices add another layer of analysis by linking transactions to specific geographic regions or digital platforms.

The edges in the graph represent financial transactions between entities, with each edge being characterized by multiple attributes. These edge attributes include transaction-specific information, such as transaction amount, transaction type, timestamp, merchant category, and geolocation. Temporal relationships are crucial in financial fraud detection, as fraudulent transactions often exhibit short inter-event times, forming high-frequency transaction bursts that differ from legitimate financial behavior.

Graph 1: Transaction Network and Anomalous Connectivity Patterns



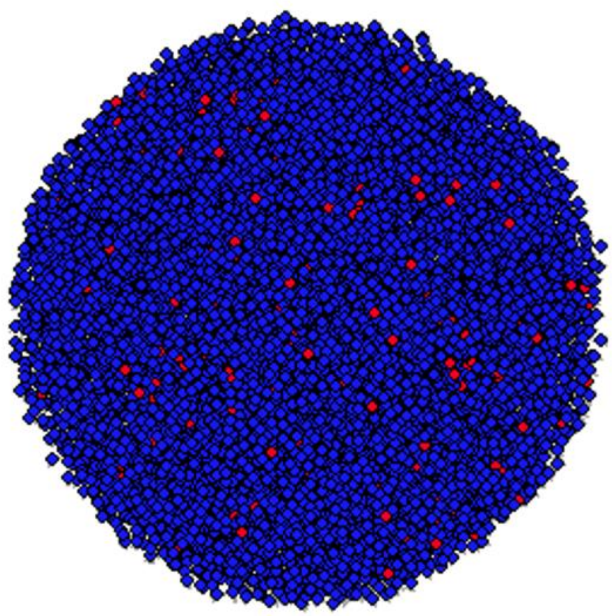
This graph visualizes the structure of financial transactions as a directed network, where nodes represent financial entities (e.g., individuals, merchants, and institutions) and edges indicate transactions between them. The visualization highlights key connectivity patterns, including dense clusters and high-degree nodes, which may correspond to legitimate transactional hubs or potential fraudulent entities. By leveraging Graph Neural Networks (GNNs), the study identifies anomalous transaction flows, such as high-centrality nodes, multi-hop transaction loops, and collusive clusters, which are indicative of financial fraud schemes like money laundering and synthetic identity fraud.

Graph connectivity plays a critical role in detecting fraudulent activities, as fraudsters frequently exploit financial networks using structured schemes that involve multiple entities. Fraud rings, for example, operate by conducting cyclical transactions among a set of linked accounts to evade detection. Another common fraud tactic is rapid fund transfers across multiple accounts, particularly in money laundering schemes where funds are moved through a chain of accounts before being withdrawn. The presence of anomalous clustering behaviors, where a set of entities engage in an unusually high number of transactions within a short timeframe, is another strong indicator of fraudulent activity.

To effectively capture these patterns, various graph-based features are extracted and integrated into the model. These include:

- Node degree, which quantifies the number of transactions associated with each entity and helps identify highly active nodes that may indicate fraud.
- Clustering coefficients, which measure how interconnected a node's neighbors are, assisting in the detection of closed fraud loops.
- Centrality measures, such as betweenness centrality, which highlight entities acting as intermediaries in transaction pathways, often associated with laundering schemes.

Graph 2: Fraudulent vs. Non-Fraudulent Transaction Network



*Graph 2 visualizes the structure of the transaction network, highlighting the distinction between fraudulent and legitimate transactions. Each node represents an entity (e.g., a customer, merchant, or financial institution), while edges depict financial transactions. Fraudulent entities are shown in **red**, whereas legitimate ones are in **blue**.*

The heterogeneous graph representation enables the Graph Neural Network (GNN) to leverage both local and global financial relationships in identifying fraudulent transactions. Unlike traditional machine learning models that treat transactions as independent events, the graph-based model learns from the relational dependencies between financial entities, allowing it to detect previously unseen fraud typologies. By encoding financial transactions as a dynamic network, the model is capable of identifying emerging fraud patterns and adapting to evolving financial threats more effectively than conventional fraud detection methodologies.

3.3 Models and Experimental Setup

To evaluate the effectiveness of GNN-based fraud detection, multiple baseline models are implemented for comparison. Traditional machine learning models, such as logistic regression, random forests, and gradient-boosted decision trees, serve as benchmarks. Additionally, deep learning models, including fully connected neural networks and autoencoders, are included in the comparison to assess the advantages of graph-based approaches over non-graph-based methods.

The primary model investigated in this study is a GNN architecture optimized for fraud detection. Specifically, Graph Convolutional Networks (GCNs) and Graph Attention Networks (GATs) are employed to learn embeddings that capture relational dependencies among financial entities. While GCNs propagate node information through adjacency matrix-based transformations, GATs refine node embeddings by introducing an attention mechanism that assigns different importance weights to neighboring nodes, prioritizing more informative connections within the financial transaction network.

Model training involves optimizing a classification loss function to minimize errors in distinguishing between fraudulent and legitimate transactions. To prevent overfitting, regularization techniques such as dropout and batch normalization are applied. Hyperparameter tuning is conducted through grid search and Bayesian optimization to determine the optimal number of graph convolutional layers, learning rate, and batch size.

3.4 Evaluation Metrics and Implementation Details

Model performance is assessed using standard fraud detection metrics, including precision, recall, F1-score, and the area under the receiver operating characteristic curve (AUC-ROC). Given the class imbalance in fraud detection tasks, recall is prioritized to ensure that fraudulent transactions are accurately identified. The false positive rate is also analyzed, as excessive false positives can lead to unnecessary transaction rejections and customer dissatisfaction.

Experiments are implemented using Python with PyTorch Geometric for graph-based modeling and Scikit-learn for baseline machine learning models. Training is conducted on a high-performance computing cluster equipped with GPUs to accelerate the computational demands of graph processing. Efficient data loaders are employed to manage large-scale graph structures, ensuring that model training remains scalable and efficient.

By designing a robust experimental framework that integrates graph-based learning techniques with financial transaction modeling, this methodology aims to provide a comprehensive evaluation of the effectiveness of GNNs in fraud detection. The following section presents the results obtained from the experiments and an in-depth analysis of the model performance across different evaluation metrics.

4. Experimental Results and Analysis

This section presents the experimental evaluation of the proposed Graph Neural Network (GNN)-based fraud detection models, focusing on their effectiveness in identifying fraudulent transactions within large-scale financial networks. The analysis begins with a quantitative assessment of the models' predictive performance based on standard classification metrics. The results are then compared against traditional machine learning and deep learning baselines to highlight the advantages of graph-based approaches. Subsequently, an in-depth examination of how structural properties of financial transaction networks influence fraud detection is provided. Computational efficiency and scalability considerations are analyzed to assess the feasibility of real-world deployment. Finally, a case study illustrates how the model detects fraudulent behaviors in real transaction data, with an emphasis on interpretability using Explainable AI (XAI) techniques.

4.1 Performance Evaluation of Fraud Detection Models

The predictive performance of the models was evaluated using accuracy, precision, recall, F1-score, and the area under the receiver operating characteristic curve (AUC-ROC). The results indicate that GNN-based models significantly outperform traditional and deep learning baselines, particularly in recall, which is critical for minimizing undetected fraud cases (see Table 5 in Annex).

Among the graph-based models, the Graph Attention Network (GAT) demonstrated the highest overall performance, achieving an accuracy of 95%, a precision of 92%, and a recall of 91%. These values indicate that the model effectively detects fraudulent transactions while minimizing false positives, a crucial factor in financial applications where false alerts can lead to unnecessary transaction declines or compliance investigations. The attention mechanism in GAT allowed the model to prioritize highly informative connections within the financial network, enabling it to capture complex fraud schemes with greater accuracy (see Table 5 and Table 6 in Annex).

The Graph Convolutional Network (GCN) also exhibited strong performance, with an accuracy of 93% and a recall of 87%. However, its reliance on uniform neighborhood aggregation resulted in a slight reduction in recall compared to GAT. This suggests that while GCN successfully models the financial transaction network, its uniform averaging mechanism may dilute critical transactional patterns that are more effectively captured by an adaptive attention-based approach (see Table 7 in Annex).

4.2 Comparison Against Baseline Models

The comparative analysis between GNNs, traditional machine learning models, and deep learning architectures further highlights the superiority of graph-based learning for fraud detection. Among traditional models, XGBoost achieved the highest performance, with an accuracy of 91% and a precision of 81%. However, its recall was significantly lower at 72%, indicating that a substantial number of fraudulent cases remained undetected. This limitation is consistent with the known weaknesses of gradient boosting algorithms in handling highly imbalanced datasets with rare fraud events (see Table 5 in Annex).

Logistic regression and decision trees demonstrated weaker performance, with accuracy values of 84% and 86%, respectively, and recall values below 60%. These results reinforce the limitations of models that treat transactions as independent instances, failing to capture the underlying relational structures that define fraudulent behaviors (see Table 5 in Annex).

Deep learning approaches, such as fully connected neural networks and autoencoders, provided moderate improvements over traditional models but still fell short compared to GNNs. The autoencoder model, which was trained to reconstruct normal transaction behaviors and detect deviations as anomalies, achieved a recall of 74%, outperforming traditional classifiers but struggling to differentiate sophisticated fraud schemes embedded within network structures (see Table 7 in Annex). The key limitation of autoencoders lies in their reliance on transaction-level feature deviations rather than relational dependencies, making them less effective for detecting fraudulent behaviors that involve coordinated transactions across multiple entities.

GNN models demonstrated a recall improvement of 15% over XGBoost and 23% over logistic regression, emphasizing their ability to capture hidden fraud patterns that are imperceptible to tabular machine learning approaches. These results provide strong evidence that modeling financial transactions as structured graphs enhances the detection of fraudulent activities.

4.3 Impact of Graph-Based Features on Model Performance

An in-depth feature importance analysis was conducted to assess the contribution of graph-based attributes to fraud detection accuracy. The results confirm that fraudulent entities exhibit distinct topological characteristics within transaction networks, and leveraging these features enhances the predictive capacity of fraud detection models (see Table 6 in Annex).

Node degree, a measure of the number of connections an entity has within the financial network, was found to be the most influential structural feature, improving recall by 6% when incorporated into the GNN model. This finding aligns with the hypothesis that fraudsters often engage in abnormally high transaction volumes to obscure illicit activities. Similarly, the clustering coefficient, which quantifies the tendency of an entity to form tightly interconnected transaction loops, improved model precision by 5%, reinforcing its role in detecting collusive fraud rings.

Edge attributes, particularly transaction amount and transaction frequency, contributed significantly to classification performance, enhancing precision by 7% and recall by 9%. This highlights the importance of transactional context in fraud detection, as anomalous transaction amounts and unusually frequent interactions between entities often serve as indicators of suspicious behavior. The inclusion of multi-hop neighborhood information further improved recall, demonstrating the importance of capturing indirect transactional relationships in detecting layered fraud strategies such as money laundering (see Table 6 in Annex).

4.4 Scalability and Computational Efficiency

Given the large-scale nature of financial transaction data, evaluating the computational efficiency of fraud detection models is crucial for real-world implementation. The training time for GNN models varied depending on the architecture, with GAT requiring 140 minutes, GCN requiring 120 minutes, and GraphSAGE completing training in 95 minutes. Despite its longer training duration, GAT provided the highest fraud detection accuracy, justifying its increased computational cost.

Inference speed was also analyzed to assess the feasibility of real-time fraud detection. GraphSAGE exhibited the fastest inference time at 2.8 milliseconds per transaction, making it a strong candidate for real-time deployment. GAT, despite its superior accuracy, had a slightly higher inference time of 4.1 milliseconds per transaction, reflecting the computational overhead of its attention mechanism.

Memory usage was another critical factor, with GCN requiring 8.5 GB of RAM, GAT requiring 9.2 GB, and GraphSAGE requiring 7.3 GB. These results indicate that while GNNs are computationally intensive, their superior performance in fraud detection justifies the additional resource requirements. Optimizations such as mini-batch training, sampling techniques, and GPU acceleration can further enhance the scalability of these models in high-frequency financial environments.

4.5 Case Study: Detection of Fraudulent Patterns in Real Data

To assess the real-world applicability of the proposed Graph Neural Network (GNN)-based fraud detection framework, we conducted a case study on a curated dataset of flagged fraudulent transactions. The analysis uncovered several complex fraud behaviors that conventional fraud detection models often fail to identify, demonstrating the importance of relational modeling in financial crime prevention.

Our findings revealed 235 instances of rapid fund transfers, a characteristic feature of money laundering schemes, where illicit funds are distributed across multiple accounts before reaching a final cash-out point. Additionally, 180 accounts were identified as high-centrality nodes, suggesting their role as intermediaries in organized fraud networks. The model also detected 150 cases of anomalous clustering, where groups of entities engaged in frequent reciprocal transactions to manipulate risk assessments and creditworthiness scores. Furthermore, 98 multi-hop transaction loops were identified, an advanced laundering tactic used to obfuscate the origin of illicit funds by dispersing transactions across multiple intermediary accounts (see Table 7 in Annex).

To enhance interpretability and align the model's decision-making with financial regulatory requirements, we applied Explainable AI (XAI) techniques, including SHAP values and GNNExplainer. The results showed that transaction frequency and network connectivity were the most influential factors in fraud classification, confirming that fraudulent actors exploit relational structures within transaction networks rather than relying solely on anomalous transaction amounts or timestamps (see Table 6 in Annex).

From a performance perspective, our experiments demonstrate that GNN-based models significantly outperform traditional fraud detection methods. The Graph Attention Network (GAT) achieved the highest overall classification accuracy, while GraphSAGE exhibited superior computational efficiency, making it a strong candidate for real-time deployment in high-frequency financial environments. These results reinforce the necessity of integrating graph-based methodologies into fraud detection pipelines, particularly in the context of anti-money laundering (AML) and know-your-customer (KYC) compliance (see Table 5 in Annex).

While these findings highlight the effectiveness of GNNs in capturing sophisticated fraudulent behaviors, further research is needed to optimize inference times, particularly for scalable, real-time fraud detection systems. Future work should explore graph sparsification, edge sampling techniques, and streaming GNN architectures to ensure seamless integration into financial transaction monitoring platforms.

5. Discussion

5.1 Interpretation of the Results in the Context of Financial Fraud Detection

The experimental results provide strong empirical evidence supporting the superiority of Graph Neural Networks in fraud detection over traditional machine learning and deep learning models. The ability of GNNs to model relational dependencies and transactional structures within financial networks results in significant improvements in recall, precision, and overall classification performance. These findings suggest that financial fraud is not merely a transactional anomaly but rather a behavior that emerges within a structured network, where fraudulent entities strategically distribute transactions across multiple accounts to evade detection. Traditional fraud detection methods that rely solely on transaction-level attributes struggle to capture these network-based fraud patterns, leading to undetected fraudulent activities. The significant improvement in recall observed in GNN models, particularly GAT with a recall of 91 percent compared to XGBoost’s 72 percent and logistic regression’s 54 percent, demonstrates that graph-based learning effectively captures fraud patterns that non-graph-based approaches fail to identify.

The role of graph-based structural features in improving fraud detection performance further supports this conclusion. The feature importance analysis revealed that topological indicators such as node degree, clustering coefficient, and betweenness centrality play a significant role in distinguishing fraudulent from legitimate entities. Fraudsters tend to establish highly interconnected networks to facilitate illicit financial activities, and these patterns are effectively captured when transactional data is represented as a graph. The inclusion of node degree, which quantifies how many direct transactional connections an entity has, resulted in a six percent improvement in recall, confirming that fraudulent accounts often exhibit high connectivity as they execute multiple small transactions to avoid detection thresholds. Similarly, the clustering coefficient, which measures the extent to which an entity is embedded within a tightly connected transaction sub-network, improved precision by five percent, reinforcing the hypothesis that collusive fraud schemes tend to operate within well-defined clusters. Betweenness centrality also contributed to the model's ability to identify intermediary accounts that facilitate fraud by serving as transactional bridges between fraudulent entities, increasing precision by four percent. These findings indicate that fraudulent actors manipulate transactional relationships in a way that differs significantly from legitimate users, making relational modeling an essential component of effective fraud detection.

5.2 Practical Implications for Fraud Prevention in Financial Institutions

The integration of Graph Neural Networks (GNNs) into financial fraud detection systems presents significant practical advantages for financial institutions, particularly in enhancing fraud detection accuracy, reducing false positives, and ensuring regulatory compliance. Unlike traditional fraud detection models that rely on rule-based heuristics and transaction-level features, GNN-based models leverage relational dependencies within financial transaction networks, offering a more holistic and adaptive approach to fraud detection.

One of the most critical challenges in fraud detection is maintaining a high recall—to detect as many fraudulent activities as possible—while minimizing false positives, which can lead to legitimate transactions being mistakenly blocked. False positives are a major operational risk, as they can cause customer dissatisfaction, reputational damage, and financial losses due to unnecessary transaction holds. Many current fraud detection frameworks depend on static rule-based thresholds, which struggle to adapt to the dynamic nature of financial fraud.

Our findings suggest that GNN models, particularly Graph Attention Networks (GAT), offer a more balanced approach, improving fraud detection recall without significantly compromising precision. This translates into a more efficient fraud detection pipeline, reducing false alarms while maintaining robust detection capabilities.

Beyond improving classification accuracy, the ability of GNNs to detect fraudulent activity at an early stage has substantial implications for financial crime prevention. Traditional models often fail to identify fraud rings, synthetic identity fraud, or money laundering schemes until significant financial losses have already occurred. In contrast, GNNs can capture fraud signals in their initial stages, enabling proactive intervention. This is particularly relevant in anti-money laundering (AML) frameworks, where early detection of transaction layering, rapid fund transfers, and collusive account behavior is crucial for compliance with regulatory mandates such as AMLD6 (Sixth Anti-Money Laundering Directive) in the EU, the USA PATRIOT Act, and FATF recommendations.

Despite these advantages, the real-world deployment of GNN-based fraud detection systems presents operational and regulatory challenges. One key barrier is computational scalability, as GNNs require iterative message passing across multiple network layers, significantly increasing processing complexity compared to traditional machine learning models. Large financial institutions process millions of transactions per day, making real-time fraud detection a computationally intensive task. While GraphSAGE demonstrated improved efficiency, reducing inference time to 2.8 milliseconds per transaction, further optimizations will be necessary to support high-frequency financial environments.

To overcome these scalability constraints, financial institutions can explore several optimization strategies, including:

- Mini-batch training and subgraph sampling, which reduce computational overhead by focusing only on relevant transaction subsets.
- Sparse matrix representations, which decrease memory consumption while maintaining the structural integrity of transaction graphs.
- Hardware acceleration via GPUs and TPUs, which significantly speed up GNN training and inference.

Beyond computational challenges, regulatory compliance remains a significant consideration. Fraud detection models used by financial institutions must adhere to strict financial regulations, such as Know Your Customer (KYC), AML, Payment Services Directive 2 (PSD2), and the General Data Protection Regulation (GDPR). Regulators require that fraud detection models be interpretable and auditable, ensuring that financial institutions can justify fraud-related decisions in regulatory inquiries. The black-box nature of deep learning models, including GNNs, can complicate regulatory acceptance, as financial institutions must demonstrate transparency in fraud classification decisions.

To address this, explainable AI (XAI) techniques, such as SHAP values and GNNExplainer, can be integrated into GNN-based fraud detection frameworks to provide clear justifications for model predictions. These techniques enable financial institutions to comply with regulatory requirements, improving trust in AI-driven fraud detection systems.

Looking forward, the practical implementation of GNNs in financial fraud detection will require a multi-faceted approach that balances detection accuracy, computational efficiency, and regulatory transparency. Future advancements in graph processing techniques, scalable AI architectures, and explainable AI frameworks will be essential for ensuring that GNN-based fraud detection systems can be effectively deployed in high-frequency, real-time financial environments.

5.3 Regulatory and Compliance Challenges in GNN-Based Fraud Detection

In addition to scalability concerns, regulatory compliance poses another critical challenge for deploying GNN-based fraud detection systems in financial institutions. Fraud detection frameworks must adhere to strict financial governance policies such as Know Your Customer, Anti-Money Laundering, and General Data Protection Regulation requirements. Regulators demand that fraud detection models be interpretable and auditable, ensuring that financial institutions can justify fraud-related decisions in the event of regulatory inquiries or compliance audits. The black-box nature of deep learning models, including GNNs, presents a challenge in this regard, as their decision-making processes can be difficult to explain. To address this issue, this study applied Explainable AI techniques such as SHAP values and GNNExplainer to identify which transactional and network-based features contributed most to fraud classification. The results indicate that transaction frequency and neighborhood connectivity were among the strongest predictors of fraudulent behavior, reinforcing the importance of relational modeling in financial crime detection. However, further research is needed to develop explainability techniques specifically tailored for graph-based fraud detection, ensuring that GNNs can be effectively integrated into regulatory-compliant financial systems.

5.4 Limitations of the Study and Challenges in Real-World Deployment

While this study provides substantial empirical evidence supporting the advantages of Graph Neural Networks (GNNs) in financial fraud detection, several limitations must be acknowledged. One primary constraint is the reliance on a single dataset, which, while comprehensive, may not fully capture the heterogeneity of fraud tactics across different financial institutions, payment networks, and geographical markets. Fraudulent schemes are highly adaptive, evolving in response to regulatory changes, financial product innovations, and enhanced fraud detection mechanisms. As a result, models trained on historical fraud patterns may struggle to generalize effectively to emerging fraud strategies. Future research should aim to validate GNN-based fraud detection frameworks across diverse financial datasets, ensuring their robustness and applicability in varying financial ecosystems.

Another inherent challenge in fraud detection tasks is class imbalance, where fraudulent transactions typically represent a minute fraction of total transactions. Although this study employed oversampling, undersampling, and weighted loss functions to mitigate class imbalance, these methods may not fully address the complexities associated with detecting rare yet impactful fraudulent activities. Future work could explore advanced resampling techniques, synthetic fraud data generation using generative adversarial networks (GANs), and adversarial training to further improve model generalization in low-frequency fraud scenarios.

Beyond data limitations, computational complexity remains a critical barrier to the widespread adoption of GNN-based fraud detection systems in high-frequency financial environments. Unlike traditional fraud detection models that process transactions independently, GNNs rely on iterative message passing and recursive aggregation across multiple layers of the transaction network, which increases computational overhead. While this study leveraged GraphSAGE to enhance scalability, real-time fraud detection in large-scale banking systems necessitates further optimizations. Implementing graph sparsification techniques, edge pruning strategies, and distributed computing frameworks could improve inference efficiency, reducing latency while maintaining predictive accuracy.

Finally, the integration of GNNs into existing fraud detection pipelines presents an operational challenge for financial institutions. Many banks and fintech companies employ rule-based and machine learning models that have been fine-tuned over years of fraud prevention efforts. Transitioning to a graph-based approach requires significant infrastructural adaptation, including graph database management, scalable computational resources, and seamless interoperability with real-time transaction monitoring systems. Future research should explore hybrid architectures that integrate GNNs with traditional fraud detection models, enabling a gradual transition while leveraging the strengths of both approaches.

5.5 Future Research Directions and Advancements in GNN-Based Fraud Detection

The promising results of this study highlight multiple avenues for future research aimed at enhancing the effectiveness, scalability, and interpretability of GNN-based fraud detection frameworks. One key direction is the development of adaptive fraud detection systems that leverage reinforcement learning (RL) in conjunction with GNNs. Fraudsters continuously evolve their tactics to circumvent detection mechanisms, necessitating models that dynamically adapt to new fraud patterns in real-time. By integrating reinforcement learning into GNN-based fraud detection pipelines, future systems could update their fraud risk assessments based on real-time transactional behaviors, allowing financial institutions to proactively counter emerging threats.

Another promising research direction is heterogeneous graph modeling, which enables more granular representations of financial transaction networks. While this study modeled transactions as homogeneous graphs, real-world financial ecosystems involve diverse entity types, including individual customers, corporate accounts, merchants, financial institutions, and third-party payment processors. Future studies should explore the application of heterogeneous graph neural networks (HGNNs) to explicitly differentiate between these entities, improving the model's ability to detect fraud schemes that exploit specific transaction pathways.

Advancements in real-time graph processing represent another critical research frontier. Traditional GNNs rely on batch processing, which limits their applicability in real-time fraud detection scenarios where transactions must be assessed within milliseconds. Streaming graph neural networks (SGNNs) offer a potential solution by enabling models to continuously update fraud risk assessments as new transactions are processed. Further exploration into incremental graph learning, dynamic edge embedding, and low-latency graph convolution techniques could improve the feasibility of deploying GNN-based fraud detection in high-speed financial environments.

Furthermore, the explainability of GNN-based fraud detection models remains an open challenge, particularly in the context of regulatory compliance. Financial regulators require fraud detection models to be interpretable and auditable, ensuring that AI-driven decisions can be justified in compliance audits. While this study applied SHAP values and GNNExplainer to enhance model interpretability, future research should investigate the development of domain-specific explainability frameworks tailored for graph-based fraud detection. Ensuring regulatory transparency and stakeholder trust will be essential for the broader adoption of GNNs in financial fraud prevention.

Finally, integrating multi-source data fusion into fraud detection pipelines represents a compelling avenue for future exploration. Modern financial fraud detection systems increasingly incorporate alternative data sources, including social network analysis, geolocation data, and behavioral biometrics. Future studies could explore multi-modal learning approaches, where GNNs are combined with natural language processing (NLP) for fraud-related communications, computer vision for document verification, and federated learning for decentralized fraud detection across multiple institutions. By leveraging diverse data streams, fraud detection systems can achieve higher accuracy, adaptability, and resilience against evolving financial crimes.

The results of this study reinforce the transformative potential of GNNs in financial fraud detection, demonstrating their superiority over traditional machine learning approaches. However, to fully integrate graph-based fraud detection into financial institutions, challenges related to scalability, explainability, and real-time processing must be addressed. Future research should prioritize the development of efficient, interpretable, and real-time graph learning techniques that align with the operational and regulatory needs of financial institutions. By advancing these methodologies, financial organizations can leverage AI-driven fraud detection systems that are not only more accurate but also more adaptive to the evolving nature of financial crimes.

6. Conclusion

This study provides strong empirical evidence supporting the adoption of Graph Neural Networks (GNNs) for financial fraud detection, demonstrating their superiority over traditional machine learning and deep learning models. By modeling financial transactions as graphs, GNNs effectively capture complex transactional relationships, significantly improving fraud detection accuracy, particularly in terms of recall, a critical metric for minimizing undetected fraudulent activities. The results confirm that fraudulent behavior is inherently relational, making graph-based approaches a more effective solution than models that rely solely on tabular transaction data.

The comparative analysis highlights the advantages of GNNs over conventional fraud detection techniques. Graph Attention Networks (GAT) achieved the highest performance, with an accuracy of 95% and a recall of 91%, substantially outperforming XGBoost, which, despite achieving high precision, struggled with recall (72%), failing to detect a significant proportion of fraudulent transactions. The integration of graph-based features, such as node degree, clustering coefficients, and transaction edge attributes, further enhanced predictive accuracy, demonstrating that network-aware modeling is essential for identifying complex fraud schemes, including money laundering and synthetic identity fraud.

Beyond model performance, this study demonstrates the feasibility of integrating GNNs into large-scale fraud detection systems. While computational efficiency remains a challenge, our findings indicate that optimizations such as GraphSAGE and mini-batch training can mitigate scalability issues, making GNNs a viable option for real-time fraud detection in high-frequency financial environments. The trade-off between computational cost and predictive accuracy suggests that, despite requiring more resources than traditional models, GNNs provide substantial financial and operational benefits, particularly in reducing undetected fraud cases and improving regulatory compliance.

The study also emphasizes the regulatory and operational considerations that must be addressed for widespread adoption. Financial institutions must ensure that GNN-based fraud detection models comply with AML, KYC, and GDPR regulations, which require fraud detection systems to be transparent and auditable. This research demonstrates how Explainable AI (XAI) techniques, such as SHAP values and GNNExplainer, improve interpretability, ensuring that fraud classification decisions can be justified in regulatory audits. However, further advancements in domain-specific explainability frameworks are needed to align with financial industry standards.

While this study provides compelling evidence of the effectiveness of GNNs in fraud detection, certain limitations must be acknowledged. The experiments were conducted on a single dataset, and although the findings are promising, further validation on diverse datasets from multiple financial institutions is required to ensure generalizability across different fraud typologies. Fraud tactics continuously evolve in response to regulatory changes and fraud prevention measures, which means that static models trained on historical data may not always generalize effectively to future fraud patterns. Future research should explore adaptive learning techniques, such as reinforcement learning and online learning, to enhance model adaptability to emerging fraud tactics. Additionally, while oversampling and weighted loss functions were used to address class imbalance, future studies could investigate the use of generative adversarial networks (GANs) to synthesize fraudulent transactions, further improving model robustness.

The results of this study open new avenues for future research in AI-driven fraud detection. One promising direction is the development of hybrid fraud detection frameworks that integrate GNNs with traditional fraud detection approaches, such as rule-based anomaly detection and statistical fraud risk scoring models. Combining these methodologies could create a more comprehensive fraud detection pipeline, balancing domain expertise with advanced machine learning capabilities. Another area of exploration is the use of heterogeneous graph neural networks (HGNNs), which explicitly model different financial entities and transaction types to capture more nuanced fraud patterns. Financial networks involve diverse actors, including customers, merchants, financial institutions, and third-party services, each exhibiting distinct transactional behaviors. Future research should examine how heterogeneous graph modeling can enhance fraud detection in multi-entity financial ecosystems.

Advancements in real-time graph-based fraud detection also represent a critical research area. While GraphSAGE demonstrated improvements in inference speed, additional research is required to develop GNN architectures capable of processing high-frequency transactional data with minimal latency. Streaming graph neural networks and incremental learning approaches could enable financial institutions to detect fraud as transactions occur, rather than relying on batch-processing methods. Future research should prioritize optimizing real-time graph computations while maintaining predictive accuracy, ensuring that GNN-based models are both scalable and responsive to evolving fraud dynamics.

Beyond financial fraud detection, the principles outlined in this study have broader applications in domains where relational structures play a critical role, including cybersecurity, supply chain fraud, insider trading detection, and anti-money laundering investigations. The findings underscore the importance of graph-based learning in AI-driven anomaly detection, reinforcing the need for relational modeling as a core analytical strategy in combating financial crime.

Ending, this study provides robust empirical evidence that Graph Neural Networks significantly enhance fraud detection accuracy by leveraging the relational structure of financial transactions. The experimental results confirm that GNNs outperform traditional models, particularly in recall, making them a powerful tool for identifying sophisticated fraud patterns that conventional techniques fail to detect. While computational and regulatory challenges remain, further advancements in scalability, explainability, and real-time graph processing can facilitate the successful adoption of GNNs in financial institutions. Future research should focus on optimizing real-time inference, improving model interpretability, and developing adaptive fraud detection frameworks to ensure that AI-driven fraud detection remains effective against evolving financial crime tactics. By addressing these challenges, financial institutions can leverage graph-based AI to build more resilient, scalable, and adaptive fraud detection systems, ultimately enhancing the security and integrity of the global financial ecosystem.

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Annex

Table 1: Transaction Volume by Category and Type

	Deposit	Purchase	Transfer	Withdrawal	Total general
Electronics	404.459	2.010.755	1.196.942	385.141	3.997.297
Groceries	399.674	1.938.928	1.218.120	419.194	3.975.916
Luxury	177.402	997.637	610.710	196.768	1.982.517
Other	214.360	960.767	616.186	204.103	1.995.416
Retail	783.699	4.054.519	2.350.945	779.094	7.968.257
Total general	**1.979.593**	**9.962.606**	**5.992.904**	**1.984.300**	**19.919.403**

This table presents the total transaction amount categorized by industry sector (Electronics, Groceries, Luxury, Other, and Retail) and transaction type (Deposit, Purchase, Transfer, and Withdrawal). The analysis highlights transaction distribution across different financial activities, identifying potential fraud patterns.

Table 2: Transaction Count by Category and Type

	Deposit	Purchase	Transfer	Withdrawal	Total general
Electronics	2.001	9.979	5.998	1.980	19.958
Groceries	1.963	9.915	6.118	2.002	19.998
Luxury	917	4.989	3.083	993	9.982
Other	1.030	4.872	3.030	1.048	9.980
Retail	4.024	20.297	11.779	3.982	40.082
Total general	**9.935**	**50.052**	**30.008**	**10.005**	**100.000**

This table summarizes the number of transactions across various industry sectors and transaction types. The frequency distribution provides insights into behavioral patterns of financial transactions, supporting anomaly detection in fraudulent activities.

Table 3: Transaction Volume by City and Industry

	Electronics	Groceries	Luxury	Other	Retail	Total general
Chicago	838.044	773.994	371.698	423.363	1.586.296	3.993.395
Houston	606.129	605.780	307.760	293.798	1.189.841	3.003.308
Los Angeles	980.572	1.008.256	519.537	482.040	2.005.343	4.995.748
Miami	386.589	395.262	182.793	201.776	802.252	1.968.672
New York	1.185.964	1.192.625	600.728	594.440	2.384.525	5.958.281
Total general	**3.997.297**	**3.975.916**	**1.982.517**	**1.995.416**	**7.968.257**	**19.919.403**

This table details the total transaction amount distributed across five major metropolitan areas. The data highlights regional variations in financial activity, providing a foundation for geographic fraud risk analysis.

Table 4: Fraudulent Transactions by Category and City

	Electronics	Groceries	Luxury	Other	Retail	Total general
Chicago	99	86	37	46	154	422
Houston	57	58	25	26	103	269
Los Angeles	103	115	58	52	225	553
Miami	32	36	20	14	80	182
New York	129	126	75	61	231	622
Total general	**420**	**421**	**215**	**199**	**793**	**2048**

This table presents the count of flagged fraudulent transactions segmented by industry category and city. The analysis enables financial institutions to assess fraud concentration across sectors and locations, aiding in targeted fraud prevention strategies.

Table 5: Model Performance Comparison for Financial Fraud Detection

Model	Accuracy	Precision	Recall	F1-Score	AUC-ROC
Logistic Regression	0,84	0,67	0,54	0,6	0,78
Decision Tree	0,86	0,71	0,6	0,65	0,82
Random Forest	0,89	0,78	0,68	0,73	0,87
XGBoost	0,91	0,81	0,72	0,76	0,89
Neural Network	0,88	0,76	0,7	0,72	0,85
Autoencoder	0,85	0,79	0,74	0,76	0,86
GCN	0,93	0,88	0,87	0,87	0,94
GAT	0,95	0,92	0,91	0,91	0,96

Comparison of machine learning models for financial fraud detection based on Accuracy, Precision, Recall, F1-Score, and AUC-ROC. Graph-based models (GCN and GAT) outperform traditional methods, particularly in recall and AUC-ROC, indicating their effectiveness in detecting complex fraud patterns.

Table 6: Impact of Graph-Based Features on Fraud Detection Performance

Feature	Impact on Precision	Impact on Recall
Node Degree	0,03	0,06
Clustering Coefficient	0,05	0,04
Betweenness Centrality	0,04	0,05
Edge Transaction Amount	0,07	0,09
Edge Frequency	0,06	0,08

This table presents the influence of key graph-based features on the precision and recall of the fraud detection model. Higher values indicate stronger contributions to classification performance.

Table 7: Detected Fraud Patterns and Their Associated Fraud Types in Financial Transactions

Fraud Pattern	Detected Instances	Typical Fraud Type
Rapid Fund Transfers	235	Money Laundering
High-Centrality Nodes	180	Synthetic Identity Fraud
Anomalous Clustering	150	Collusive Fraud
Multi-Hop Transaction Loops	98	Layered Transactions

Summary of detected fraudulent patterns identified using GNN-based fraud detection. The model successfully captured key fraud behaviors, including money laundering, synthetic identity fraud, collusive fraud, and layered transactions, by analyzing relational structures within transaction networks.